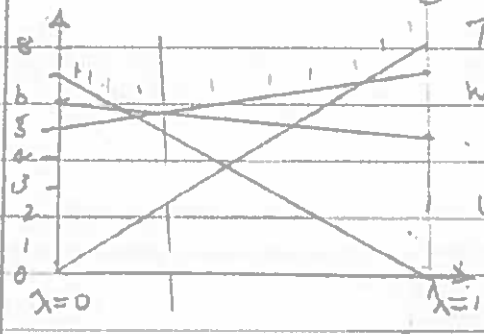
# Model Answers to 2017 Game Theory Exam

1 (i) The pay-off matrix is $\begin{matrix} & B \\ A & \end{matrix}\begin{bmatrix} 6 & 5 & 7 & 0 \\ 5 & 7 & 0 & 8 \end{bmatrix}$

To calculate a safety strategy for A "=" $(1-\lambda)\times$ row (1) $+ \lambda \times$ row (2) we look at the diagram:




The worst case pay-off (for A) is minimized when A chooses $\lambda$ to satisfy

$6(1-\lambda) + 5\lambda = 5(1-\lambda) + 7\lambda$

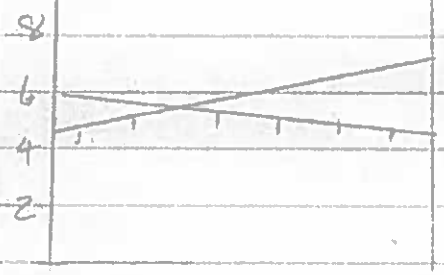
with solution $\lambda = \frac{1}{3}$.

So A's mixed strategy is

$\frac{2}{3} \times$ row (1) $+ \frac{1}{3} \times$ row 2

Because, when A chooses a safety strategy, it will not be favorable for B to choose columns (3) or (4), we can ignore these. The pay-off matrix then becomes

$\begin{bmatrix} 6 & 5 \\ 5 & 7 \end{bmatrix}$




The worst case pay-off (for B) is maximized when B chooses $\mu$ to satisfy

$6(1-\mu) + 5\mu = 5(1-\mu) + 7\mu$

with solution $\mu = \frac{1}{3}$.

So A's mixed strategy is $\frac{2}{3} \times$ row (1) $+ \frac{1}{3} \times$ row (2) $+ 0 \times$ row (3) $+ 0 \times$ row (4)

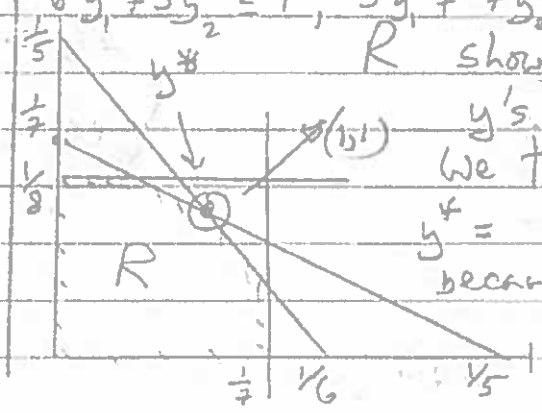
The pay-off is $(\frac{2}{3}, \frac{1}{3}) \begin{bmatrix} 6 & 5 & 7 & 0 \\ 5 & 7 & 0 & 8 \end{bmatrix} \begin{pmatrix} 2/3 \\ 1/3 \\ 0 \\ 0 \end{pmatrix} = \frac{17}{3} = V$

The equivalent LP giving A's safety strategy is:

$$\begin{cases} \text{Max } y_1 + y_2 \\ \begin{bmatrix} 6 & 5 \\ 5 & 7 \\ 7 & 0 \\ 0 & 8 \end{bmatrix} \leq 1 \\ y_1 \geq 0, \; y_2 \geq 0 \end{cases}$$

The region in 2-space of $y$'s satisfying the constraints is defined by the inequalities:

$6y_1 + 5y_2 \leq 1, \; 5y_1 + 7y_2 \leq 1, \; 7y_1 \leq 1, \; 8y_2 \leq 1, \; y_1 \geq 0, \; y_2 \geq 0$.



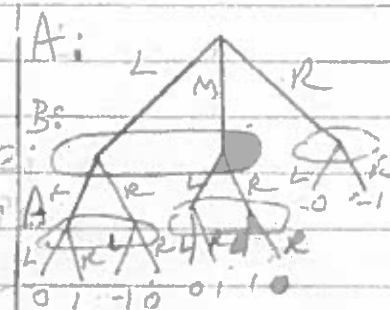

R shows the 'feasible' region for $y$'s.

We find (by solving $6y_1 + 5y_2 = 1$, $5y_1 + 7y_2 = 1$) $y^* = (\frac{2}{17}, \frac{1}{17})$. This solves the (LP) because it is furthest feasible point in the (1,1) direction. We see

$y^* = \frac{1}{V} \times (\frac{2}{3}, \frac{1}{3})$

# 2017 Game Theory Exam

2.




Because Player A remembers his first action the strategies for Player A are

$A1 = \binom{L}{L}$, $A2 = \binom{L}{R}$, $A3 = \binom{M}{L}$, $A4 = \binom{M}{R}$, $(A5) = \binom{R}{L}$, $A6 = \binom{R}{R}$

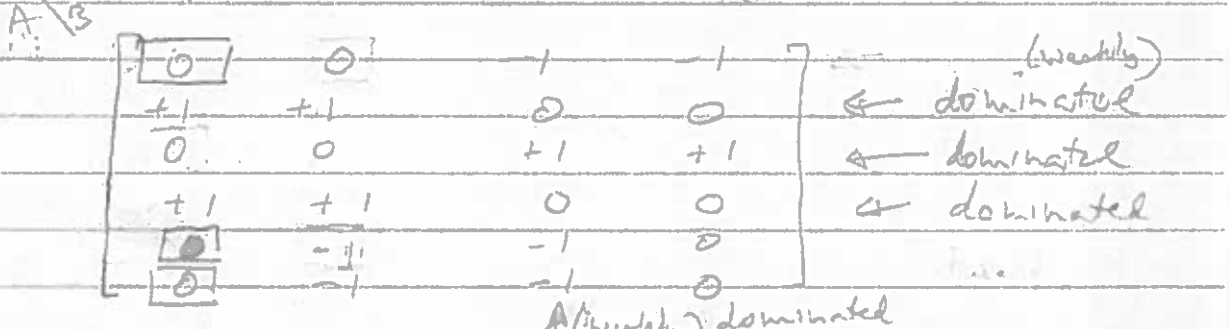
For the given B-player information sets, the strategies for Player B are

$$B1 = \begin{cases} L & \text{if } A = L \text{ or } M \\ L & \text{if } A = R \end{cases}, \quad B2 = \begin{cases} L & \text{if } A = L \text{ or } M \\ R & \text{if } A = R \end{cases}$$

$$B3 = \begin{cases} R & \text{if } A = L \text{ or } M \\ R & \text{if } A = R \end{cases} \quad B4 = \begin{cases} R & \text{if } A = L \text{ or } M \\ L & \text{if } A = R \end{cases}$$

The pay-offs for the different choices are

| A \ B | B1 | B2 | B3 | B4 |
|---|---|---|---|---|
| A1 | LLL=0 | LLL=0 | LRL=-1 | LRL=-1 |
| A2 | LLR=1 | LLR=1 | LRR=0 | LRR=0 |
| A3 | MLL=0 | MLL=0 | MRL=1 | MRL=1 |
| A4 | MLR=1 | MLR=1 | MRR=0 | MRR=0 |
| A5 | RLL=0 | RRL=-1 | RRL=-1 | RLL=0 |
| A6 | RLR=0 | RRR=-1 | RRR=-1 | RLR=0 |




rows 2, 3 and 4 are weakly dominated (by row 1)
column 3 is weakly dominated (by column 4)

There are 3 Nash equilibria (in pure strategies):
(A1, B1) = LLL, (A5, B1) = RLL, (A6, B1) = RLR.
All these Nash equilibria are admissible.

# 2017 Game Theory Exam

3. Consider the following events

(a) $A_1$ : {Player 1 dials up and Players 2, .., N dials up}

$A_2$ : {Player 1 dials up and at least one of the players 2, .., N does not dial up}

$A_3$ : {Player 1 does not dial up}

The satisfaction levels for player 1 are:

satisfaction level = −1 if $A_1$ occurs

" " = +3 if $A_2$ occurs

" " = 0 if $A_3$ occurs

Let $(\bar{\beta}, \ldots, \bar{\beta})$ be a symmetric collection of probabilities for the N-players. We assume that $0 < \bar{\beta} < 1$. Take another, arbitrary, probability $\beta$ for the 1st player. Then, for the first player

$$\text{Pay off}(\beta, \bar{\beta}, \ldots, \bar{\beta}) \leq \text{Pay off}(\bar{\beta}, \ldots, \bar{\beta}) \quad \text{— (*)}$$

since $(\bar{\beta}, \ldots, \bar{\beta})$ is a Nash equilibrium. But

$$\text{Pay off}(\beta, \bar{\beta}, \ldots, \bar{\beta}) = (-1)\,\text{Prob}\{A_1\} + 3\,\text{Prob}\{A_2\} + 0\,\text{Prob}\{A_3\}$$

$$\text{Prob}\{A_1\} = \text{Prob}\{\text{Player 1 dials up}\} \times \text{Prob}\{\text{Player 2..N did all dial up}\}$$

(by independence)

$$= \beta \times \bar{\beta}^{N-1}$$

$$\text{Prob}\{A_2\} = \text{Prob}\{\text{Player 1 dials up}\} \times \left(1 - \text{Prob}\{\text{Prob 2..N all dial up}\}\right)$$

$$= \beta \times (1 - \bar{\beta}^{N-1})$$

$$\text{Prob}\{A_3\} = \text{Prob}\{\text{Play 1 does not dial up}\} = (1-\beta)$$

So pay-off is $-1 \times \beta \times \bar{\beta}^{N-1} + 3 \times \beta(1 - \bar{\beta}^{N-1}) + 0 \times \beta$

$$= (-\bar{\beta}^{N-1} + 3(1 - \bar{\beta}^{N-1}))\beta$$

If this is maximized over $\beta$ at $\bar{\beta}$ (see (*)), the slope

$$-\bar{\beta}^{N-1} + 3(1 - \bar{\beta}^{N-1}) = 0$$

$$\Rightarrow \quad 4\bar{\beta}^{N-1} = 3, \quad \text{So } \bar{\beta} = \left(\tfrac{3}{4}\right)^{1/(N-1)}$$

(b) Communication fails when all subscribers dial in.

The probability $= \bar{\beta}^N = \bar{\beta} \times \bar{\beta}^{N-1} = \tfrac{3}{4}\left(\tfrac{3}{4}\right)^{1/N-1}$.



# 2017 Game Theory Exam

4A(i) $L^X(x,a,b) = -a-b+x^2$, $L^A(x,a,b) = ab - xa + a^2 - \frac{1}{2}b^2$
and $L^B(x,a,b) = ab - 2xb + b^2 - \frac{1}{4}a^2$

Fix $x$. Then for any $a$, $L^A(x,a,\cdot)$ is maximized when $\partial L^A/\partial b \; L^A(x,a,b) = 0 \Rightarrow a - b = 0 \Rightarrow b = a$

A will choose $a$ to satisfy $\partial/\partial a \; L^A(x,a,b=a) = 0$
$\Rightarrow \frac{\partial}{\partial a}\{a^2 - xa + a^2 - \frac{1}{2}a^2\} = 0 \Rightarrow 3a - x = 0$ or $a = \frac{1}{3}x$.

This is A's safety strategy.

Likewise, B's safety strategy is obtained by solving
$\partial/\partial a \; L^B(x,a,b) = 0 \Rightarrow b - \frac{1}{2}a \Rightarrow a = 2b$
and $\partial/\partial b \; L^B(x,a=2b,b) = \frac{\partial}{\partial b}\{2b^2 - 2xb + b^2 - b^2\} = 0$
$\Rightarrow 4b - 2x = 0$ or $b = x/2$.

X's pay-off is $L^X(x, a=\frac{1}{3}x, b=\frac{x}{2}) = -\frac{1}{3}x - \frac{x}{2} + x^2$
This is maximized when $2x = 5/6 \Rightarrow X = \frac{5}{12}$

The combined strategies of X, A, B are
$\left(x = \frac{5}{12},\ a = \frac{5}{36},\ b = \frac{5}{24}\right)$

A(ii) Now assume A and B choose $a$ and $b$ to give a Nash equi
Then $\frac{\partial}{\partial a} L^A(x,a,b) = 0$, $\partial/\partial b \; L^B(x,a,b) = 0$
$\Rightarrow b - x + 2a = 0$ and $a - 2x + 2b = 0$
These can be solved to give $a = 0$, $b = x$
X's pay-off is $L^X(x, a=0, b=x) = -x + x^2$
This is maximized when $2x = 1 \Rightarrow x = \frac{1}{2}$
Now the strategies are: $\left(x = \frac{1}{2},\ a = 0,\ b = \frac{1}{2}\right)$

B The safety strategy $\bar{a}$ of A minimizes $\max_{b \in B}\{L^A(a,b)\}$
By definition of $\phi(\cdot)$, $\max_{b' \in B} L^A(\bar{a},b') = L^A(\bar{a},\bar{b})$
Now take any $a \in A$. Then $\max_{b' \in B} L^A(a,b') = L^A(a,\phi(a))$
But, by the tangency condition, $(a,\phi(a))$, which is on the inverted response curve, lies outside the region $R$.
So $L^A(\bar{a},\bar{b}) < L^A(a,\phi(a))$. Hence $\bar{a}$ is a safety strategy.

# 2017 Game Theory Exam

5 (i) The dynamic programming approach is to solve for $V_0(x), \ldots V_N(x)$

$$\begin{cases} V_t(x) = \min_u \max_v \{ V_{t+1}(f(x,u,v)) + L(x) \} & t=0,\ldots,N-1 \quad (1) \\ V_N(x) = L(x) \end{cases}$$

We assume a "saddle point" $(\bar{u}_t(x), \bar{v}_t(x))$ exists for each $t=0,\ldots,N-1$ and relevant values of $x$.
Then $\{(\bar{u}_t(x), \bar{v}_t(x))\}$ is a saddle point for the dynamic game, and $V_0(x_0)$ is the value of the game.

(ii) For the data as given, $V_N(x) = k_N x + c_N$ with $k_N = 1$, $c_N = 0$
To confirm $V_t(x)$ has 'linear structure' for all $t$, assume
$V_{t+1}(x) = k_{t+1} x + c_{t+1}$ ( for some $k_{t+1} > 0$, some $c_{t+1}$ )
The right side of (1) is

$$(k_{t+1}+1)x + c_{t+1} + k_{t+1} \times \left( \min_u \max_v u^T S(x) v \right) \quad (2)$$

$\min_u \max_v u^T S(x) v = \min_u \max_v (u_1, u_2) \begin{bmatrix} 2 & 1 \\ 1 & x \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$
i.e. the pay-off of a zero sum game in mixed strategies.
The saddle strategies $(\bar{u}, \bar{v})$ are $((1-\mu, \mu), (1-\nu, \nu))$ where
$(1-\mu)2 + \mu = (1-\mu)1 + x\mu \Rightarrow \bar{u} = \left(\frac{x-1}{x}, \frac{1}{x}\right)$
$(1-\nu)2 + \nu = (1-\nu)1 + x\nu \Rightarrow \bar{v} = \left(\frac{x-1}{x}, \frac{1}{x}\right)$
The pay-off is

$$\bar{u}^T S(x) \bar{v} = \frac{1}{x^2}\left(2(x-1)^2 + 2(x-1) + x\right) = \frac{(2x-1)x}{x^3}$$

So (2) becomes

$$(k_{t+1}+1)x + c_{t+1} + k_{t+1} + \frac{2x-1}{x}$$
$$= (3k_{t+1}+1)x + c_{t+1} - k_{t+1}$$

So $V_t(x) = k_t x + c_t$, where $k_t = 3k_{t+1} + 1$, $c_t = c_{t+1} - k_{t+1}$
Summary: Calculate $k_t, c_t$, $t=0,\ldots,N-1$ from

$$\begin{cases} k_t = 3k_{t+1} + 1 & \text{with final values } k_N = 1,\ c_N = 0 \\ c_t = c_{t+1} - k_t \end{cases}$$

Then the value of the game is $k_0 x_0 + c_0$ $(= V_0(x))$
The saddle point strategies are

$$(\bar{u}(x), \bar{v}(x)) = \left(\left(\frac{x-1}{x}, \frac{1}{x}\right), \left(\frac{x-1}{x}, \frac{1}{x}\right)\right) \quad (x > 1)$$